

THIRD SEMESTER (CBCSS-UG) DEGREE EXAMINATION, NOVEMBER 2020

Mathematics

MTS 3C 03—MATHEMATICS – 3

Time : Two Hours

Maximum : 60 Marks

Section A

*Answer at least eight questions.**Each question carries 3 marks.**All questions can be attended.**Overall Ceiling 24.*

1. If $\vec{r}(t) = 2\cos t \hat{i} + 6\sin t \hat{j}$, find $\frac{d\vec{r}}{dt}$ at $t = \frac{\pi}{2}$.
2. Find the curvature of a circle whose radius is 2.
3. If $z = e^x \sin(xy)$, find $\frac{\partial^2 z}{\partial y^2}$.
4. Find the gradient of $f(x, y, z) = xy^2 + 3x^2 - z^3$ at $(1, 1, 1)$.
5. Show that $\text{div } \vec{r} = 3$.
6. Evaluate $\int_2^4 \int_1^3 (40 - 2xy) dx dy$.
7. Use double integrals to find the area of the plane region enclosed by the curves $y = \sin x$ and $y = \cos x$ for $0 \leq x \leq \frac{\pi}{4}$.
8. Find the Jacobian of $u = \frac{y}{x^2}$, $v = xy$.
9. Sketch the graph of the region $|z - 2i| = 2$.
10. Write the real and imaginary part of $f(z) = \sin z$.

Turn over

11. Evaluate $\oint_C \frac{z^2}{z-1} dz$, where C is $|z| = 2$.

12. Evaluate $\int_C z dz$ where C is given by $x = t^2, y = t$ from $0 \leq t \leq 1$.

(8 × 3 = 24 marks)

Section B

Answer at least five questions.

Each question carries 5 marks.

All questions can be attended.

Overall Ceiling 25.

13. Find the directional derivative of $F(x, y, z) = xy^2 - 4x^2y + z^2$ at $(1, -1, 2)$ in the direction of $6\hat{i} + 2\hat{j} + 3\hat{k}$.

14. Find an equation of the tangent plane to the graph of $x^2 - 4y^2 + z^2 = 16$ at $(2, 1, 4)$.

15. Use Green's theorem to evaluate $\oint_C (x^2 - y^2) dx + (2y - x) dy$, where C consists of the boundary of the region in the first quadrant that is bounded by $y = x^2$ and $y = x^3$.

16. Change the order of integration and hence evaluate $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dy dx$.

17. Use divergence theorem to evaluate $\iiint_S (\vec{F} \cdot \hat{n}) dS$ where $\vec{F} = xy\hat{i} + y^2z\hat{j} + z^3\hat{k}$ and S is the unit cube defined by $0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1$.

18. Evaluate $\oint_C \left(z + \frac{1}{z} \right) dz$, where C is the unit circle $|z| = 1$.

19. Evaluate $\oint_C \frac{z^4 - 3z^2 + 6}{(z+i)^3} dz$, where C is $|z| = 2$.

(5 × 5 = 25 marks)

Section C

*Answer any one question.
The question carries 11 marks.*

20. Use Stoke's theorem to evaluate $\oint_C z \, dx + x \, dy + y \, dz$, where C is the trace of the cylinder $x^2 + y^2 = 1$ in the plane $y + z = 2$ counter clockwise as viewed from above.
21. Find the volume of the solid in the first octant bounded by the graphs of $z = 1 - y^2$, $y = 2x$ and $x = 3$.

(1 × 11 = 11 marks)

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